

ASTEROIDS

Mainly moving between the orbits of Mars and Jupiter, asteroids are the remnants of a planet-formation process that failed. Today's asteroid belt contains only about 100 asteroids that are larger than 125 miles (200 km) across. But there are 100,000 asteroids greater than about 12.5 miles (20 km) across and a staggering 1 billion that are over 1.25 miles (2 km) along their longest axis. Ceres, the first asteroid to be discovered, in 1801, is now also called a dwarf planet (see p.175). Ceres contains about 25 percent of the mass of all the asteroids combined.

EROS

Only asteroids bigger than about 215 miles (350 km) in diameter are spherical. Eros is an irregularly shaped fragment of a much larger body.



MAIN-BELT ASTEROID

951 Gaspra

AVERAGE DISTANCE TO SUN 206 million miles (331 million km)

ORBITAL PERIOD 3.29 years

ROTATION PERIOD 7.04 hours

LENGTH 11.2 miles (18 km)

DATE OF DISCOVERY July 30, 1916

Until 1991, asteroids could be glimpsed only from afar. In October of that year, a much closer view was obtained when the Galileo spacecraft flew within 1,000 miles (1,600 km) of Gaspra, taking 57 color images. Gaspra is a silicate-rich asteroid. The surface is very gray, with some of the recently exposed crater edges being bluish and some of the older, low-lying areas appearing slightly red.



IRREGULAR SHAPE

MAIN-BELT ASTEROID

5535 Annefrank

AVERAGE DISTANCE TO SUN 206 million miles (331 million km)

ORBITAL PERIOD 3.29 years

ROTATION PERIOD Not known

LENGTH 3.7 miles (6 km)

DATE OF DISCOVERY March 23, 1942

Annefrank orbits in the inner regions of the Main Belt of asteroids and is a member of the Augusta family. On November 2, 2002, Annefrank was imaged by NASA's Stardust spacecraft as it passed within 2,050 miles (3,300 km) on its way to Comet Wild 2. Annefrank turned out to be twice as large as had been predicted from Earth-based observations. The brightness that is detected from Earth is proportional to the reflectivity multiplied by the surface area,

but astronomers had used too high a value for the reflectivity. The asteroid was named after the famous diarist Anne Frank, who died during the Holocaust.

SURFACE BRIGHTNESS

This image of Annefrank, taken by the Stardust probe in 2002, shows differences in brightness over the surface of the asteroid. The variations are mainly due to dusty soil layers reflecting different amounts of sunlight in different directions.



NEAR-EARTH ASTEROID

4179 Toutatis

AVERAGE DISTANCE TO SUN 234 million miles (376 million km)

ORBITAL PERIOD 4.03 years

ROTATION PERIOD 5.4 and 7.3 days

LENGTH 2.65 miles (4.26 km)

DATE OF DISCOVERY January 4, 1989

Toutatis was named after a Celtic god (who, incidentally, appears in the *Asterix* comic books). A typical near-Earth asteroid, it sweeps past the planet nearly every 4 years. In September 2004, it came as close as just four times the distance of the Earth to the Moon. Toutatis is an S-class asteroid, similar to a stony-iron meteorite in composition. It tumbles in space much like a football after a botched pass, spinning around two axes, with periods of 5.4 and 7.3 days.



RADAR IMAGE

MAIN-BELT ASTEROID

2867 Šteins

AVERAGE DISTANCE TO SUN 220 million miles (354 million km)

ORBITAL PERIOD 3.63 years

ROTATION PERIOD 6.05 hours

DIAMETER 4.14 miles (6.67 km)

DATE OF DISCOVERY November 4, 1969

Some asteroids are not solid but consist of rock fragments with gaps between. One such example—termed a “rubble pile”—is Šteins, which the Rosetta space probe showed to be shaped like a cut diamond. The impact that produced its largest crater is thought to have fractured the asteroid.



DIAMOND IN THE SKY

MAIN-BELT ASTEROID

21 Lutetia

AVERAGE DISTANCE TO SUN 227 million miles (365 million km)

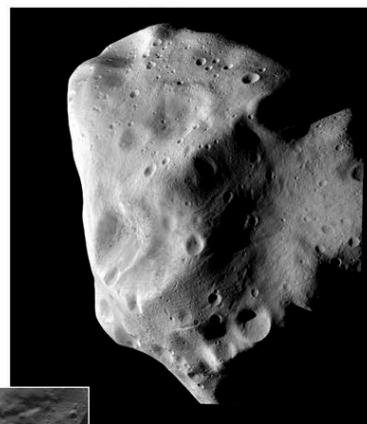
ORBITAL PERIOD 3.80 years

ROTATION PERIOD 8.17 hours

DIAMETER 75 miles (121 km)

DATE OF DISCOVERY November 15, 1852

Lutetia was the second of two asteroids visited by the European Space Agency's Rosetta space probe, the first being 2867 Šteins (see left). At over 60 miles (100 km) across, Lutetia is one of the larger asteroids and is also one of the most dense, suggesting that it contains large amounts of iron and might once have had a molten core. Rosetta's images, taken in July 2010, showed hundreds of craters up to 34 miles (55 km) wide and boulders as large as 1,000 ft (300 m) across on Lutetia's battered surface. Lutetia may have been almost spherical before parts of it were chipped off.



ROSETTA'S VIEW

Lutetia was photographed by Rosetta at its closest approach of 1,970 miles (3,170 km) in the top image. Craters and grooves on the asteroid's surface are visible in the close-up (left).



Some of its craters have been partly or completely buried by landslides set off by the vibrations from later impacts. Lutetia seems to be a link between small “rubble pile” asteroids and terrestrial planets such as Earth.

MAIN-BELT ASTEROID

253 Mathilde

AVERAGE DISTANCE TO SUN 246 million miles (396 million km)

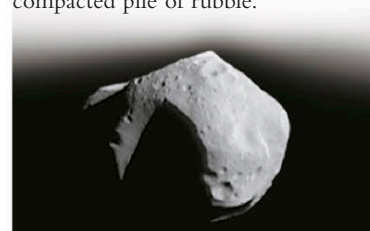
ORBITAL PERIOD 4.31 years

ROTATION PERIOD About 418 hours

LENGTH 41 miles (66 km)

DATE OF DISCOVERY November 12, 1885

Mathilde was visited by the NEAR Shoemaker space probe in 1997 but, because it spins very slowly, only about half of the surface was imaged. It is a primitive carbonaceous asteroid with a density much lower than that of most rocks, suggesting that it is full of holes. Mathilde is probably a compacted pile of rubble.



WEDGE-SHAPED CRATER